Southern New Hampshire University

CS 255 Business Requirements Document Template

System Components and Design

Purpose

What is the purpose of this project? Who is the client and what do they want their system to be able to do?

- The purpose of this project is to design a cloud-based web system for DriverPass that helps students better prepare for driving tests through lessons, tests, and up-to-date DMV policies.

- The client is DriverPass which is owned by Liam. They want a system that allows students to take online classes and practice tests, schedule and manage on-the-road driving lessons, and allows staff to manage reservations, packages, user accounts, and the student's training progress.

- The system must support online access from any computer or mobile device for ease of access, role-based security, activity tracking, package management, and integration with DMV updates, so training materials stay current.

System Background

What does DriverPass want the system to do? What is the problem they want to fix? What are the different components needed for this system?

- DriverPass wants to solve the societal problem of high failure rates when it comes to DMV driving tests, caused by inadequate student preparation.

- Their idea of a solution is a combination of online practice exams and classes as well as supervised on-the-road training offered through three flexible packages.

- Key system components include:

  -Online student registration and profile management (name, address, phone, email, credit-card information, pick-up/drop-off location).

  -Reservation system for two-hour driving lessons that matches students with drivers and cars.

  -Support for 3 separate training packages (6-hour, 8-hour + classroom lesson, 12-hour + classroom lesson + online class access) with the ability to disable packages.

  -Online class content, practice tests, and student progress tracking (test name, time, score, status).

  -Driver notes and lesson records (lesson time, start/end hour, comments).

  -Role-based access for Owner, IT Supervisor, Secretary, Customers, and Drivers.

-Activity logging and printable reports for accountability.

-Cloud hosting so DriverPass does not have to manage backup or security infrastructure.

-Notification mechanism for DMV rule/policy updates.

Objectives and Goals

What should this system be able to do when it is completed? What measurable tasks need to be included in the system design to achieve this?

- It should provide secure online access to system data from any computer or mobile device and allow users to download or print reports for offline use.
- Support distinct user roles with appropriate permissions and give the IT Supervisor full administrative rights to reset passwords and block accounts
- Allow customers to create, modify, and cancel driving-lesson reservations online or through the secretary by phone or in the office
- Track and display which student is assigned to which driver, car, date, and time for every lesson.
- Capture complete customer registration information and store the pick-up/drop-off location
- Deliver online class materials, practice tests, and display each student's test progress (name, time taken, score, status: not taken / in progress / failed / passed).
- Record and display the driver's notes for each lesson in a structured format (lesson time, start hour, end hour, comments).
- Enable authorized staff to disable any training package so new customers cannot register for it
- Maintain a connection or notification process with the DMV, so rules, policies, and sample questions can be updated/current, and staff are notified of changes.
- Generate printable activity reports that show who created, modified, or canceled each reservation (tracking)
- Host the entire application in the cloud so that DriverPass does not have to manage servers, backups, or security infrastructure

Requirements

Nonfunctional Requirements

In this section, you will detail the different nonfunctional requirements for the DriverPass system. You will need to think about the different things that the system needs to function properly.

Performance Requirements

What environments (web-based, application, etc.) does this system need to run in? How fast should the system run? How often should the system be updated?

- The system should be web-based and be available to access from both desktop and mobile devices.
- The page load time should preferably be under 3 seconds under a normal load
- The system should be able to handle numerous concurrent users (students, staff)
- Online practice tests and class content should load quickly and be responsive during use
- The system should support near real-time or real-time updates for status and status changes

Platform Constraints

What platforms (Windows, Unix, etc.) should the system run on? Does the back end require any tools, such as a database, to support this application?

- The system should be cloud-based, which is preferred by DriverPass so that they do not have to manage servers, or security infrastructure
- The backend will require a relational database to store customer data, packages, reservations, activity logs, and student progress
- The system should run on standard web technologies and be accessible from any modern operating system (Windows, iOS, Android)
- The system should provide offline access which should be limited to downloading reports

Accuracy and Precision

How will you distinguish between different users? Is the input case-sensitive? When should the system inform the admin of a problem?

- The system should identify uniquely each user with a combination of a username and password, along with a role
- Usernames and email addresses can be case-insensitive for login
- Events such as reservations, cancellations, and modifications should be accurately logged with a user ID, a time stamp, and an action type
- Credit-card and personal information must be validated for the correct format before being stored
- The system should notify the Administrator in the case of critical errors, failed login attempts beyond a set amount, or data inconsistencies

Adaptability

Can you make changes to the user (add/remove/modify) without changing code? How will the system adapt to platform updates? What type of access does the IT admin need?

- Authorized staff (Owner or IT Supervisor) shall be to disable a training package without require code changes
- The system should support future addition or removal of packages through configuration rather than hard-coding.
- Role permissions shall be adjustable so that access rights can be adjusted without rewriting the application
- The system should be designed to adapt or accommodate to platform or browser updates with minimal disruption
- The IT Supervisor should have full administrative access to users, access to reset passwords, and block accounts.

Security

What is required for the user to log in? How can you secure the connection or the data exchange between the client and the server? What should happen to the account if there is a "brute force" hacking attempt? What happens if the user forgets their password?

- Users should log in with a unique username or email, and password.
- All of the data between the client and the server should be encrypted (HTML/TLS)
- After a defined number of failed log in attempts, the account should be temporarily locked, and an IT Supervisor should be notified
- Customers should be able to request an automatic password reset via email if desired
- Sensitive data such as credit card information and personal details should be stored securely and only be accessible to authorized roles

Functional Requirements

Using the information from the scenario, think about the different functions the system needs to provide. Each of your bullets should start with "The system shall . . ." For example, one functional requirement might be, "The system shall validate user credentials when logging in."

- The system shall allow customers to register online as well as by calling the secretary (or in person) to give information such as first name, last name, phone number, address, state, credit-card information, and pick-up/drop-off location.
- The system shall allow customers to create, modify, and cancel driving-lesson reservations online or through the secretary
- The system shall match each registration with a specific driver, car, time, and date
- The system shall support three predefined training packages and allow staff to disable packages so new customers cannot register for it
- The system shall provide online class content including practice tests for eligible packages and track each student's progress (test name, time taken, score, status)

- The system shall allow drivers to record structured notes for each lesson (lesson time, start time, end time, comments)
- The system shall maintain a complete activity log of who created, modified, or cancelled a reservation and generate printable activity reports, available for offline download
- The system shall support role-based access for Owner, IT Supervisor, Secretary, Customers, and Drivers.
- The system shall allow the IT Supervisor to reset passwords and block user accounts
- The system shall provide a mechanism to receive or be notified upon DMV rule, policy, or sample-question updates so training materials remain current

User Interface

What are the needs of the interface? Who are the different users for this interface? What will each user need to be able to do through the interface? How will the user interact with the interface (mobile, browser, etc.)?

- The interface shall be web-based and accessed from any modern browser on desktop or mobile device
- Different user roles such as Owner, IT Supervisor, Secretary and Customer shall see role-based menus and functions after logging in
- Customers shall be able to register, view current packages, schedule/modify/cancel lessons, access online practice tests and classes, and view their progress through a clean, intuitive, responsive interface
- The Secretary shall have an appropriate interface for creating and managing reservations on behalf of customers by both call and walk-in
- Drivers shall have a basic interface to view their assigned lessons, lesson notes, and customers.
- The Owner and IT Supervisor shall have administrative dashboards for managing packages, users, activity reports, as well as system settings
- The interface shall follow the visual direction that was indicated by Liam of DriverPass (clean, professional, easy to navigate)
- All critical actions such as reservations, cancellations, and package changes shall provide clear confirmation messages

Assumptions

What things were not specifically addressed in your design above? What assumptions are you making in your design about the users or the technology they have?

- Customers and staff have access to reliable internet when they need to modify data
- DriverPass will be providing the content for the practice tests and online classes
- The three packages mentioned will remain the initial set of packages.
- A mechanism or agreement with the DMV for receiving notifications upon role or policy update
- Drivers and cars will be pre-assigned or managed outside the core reservation matching logic in the initial version
- Credit-card processing will be handled by third-party payment gateway (allowing the system to only have to store the necessary information or token)

Limitations

Any system you build will naturally have limitations. What limitations do you see in your system design? What limitations do you have as far as resources, time, budget, or technology?

- True offline data modification is not supported. Only download for offline viewing is available
- Full ability to have non-technical staff create or delete packages is deferred to a future release. Only package disabling for now
- The system depends on cloud-provider uptime and internet connectivity for all real-time operations
- The initial version focuses on the core reservation, package, progress-tracking, and activity logging functions. Advanced analytics or mobile native apps would be out of scope
- If a direct API is not available, integration with the DMV for automatic updates may be limited to notifications or manual import

Gantt Chart

Please include a screenshot of the GANTT chart that you created with Lucidchart. Be sure to check that it meets the plan described by the characters in the interview.

